Artificial Intelligence and Expert Systems
CSE - 321

Genetic Algorithm

Contents

- Evolution
- Natural Evolution
- Evolutionary Computation
- Genetic Algorithms
- Evolution Strategy
- Genetic Programming
- An Example Scenario

Evolution?

- Evolution: A series of genetic changes by which a living organism acquires characteristics that distinguish it from other organisms.
- Human are the products of evolution.
- So, by modeling the process of evolution, we might expect to create intelligent behaviour.

Natural Evolution

- On 1st July 1858, Darwin presented his theory of evolution
- Darwin's classical theory of evolution, together with Weisman's theory of natural selection and Mendel's concept of genetics now represent the Neo-Darwinian paradigm.
- Neo-Darwinism is based on the process of reproduction, mutation, competition and selection.

Evolutionary Computation

- Evolutionary computation simulates evolution on a computer.
- The evolution approach to machine learning is based on computational model of natural selection and genetics. We call them evolutionary computation, an umbrella term that combines genetic algorithms, evolution strategies and genetic programming.

Genetic Algorithms

- A type of evolutionary computation inspired by Darwin's theory of evolution.
- A genetic algorithm generates a population of possible solutions encoded as chromosomes, evaluates their fitness, and create a new population by applying genetic operators-crossover and mutation.
- By repeating this process over many generations, the genetic algorithm breeds an optimal solution to the problem.

Keywords Related to GA

- **Gene:** A basic unit of chromosome that controls the development of a particular feature of a living organism. In Holland's (Founder of GA) Chromosome a gene is represented by either 0 or 1.
- **Chromosomes:** A string of genes that represent an individual. Each chromosome consists of a number of gene.
- **Fitness:** The ability of a living organism to survive and reproduce in a specific environment.
- **Crossover:** A reproduction operator that creates a new chromosome by exchanging parts of two existing chromosome.
- **Mutation:** A genetic operator that randomly change the gene value in a chromosome.

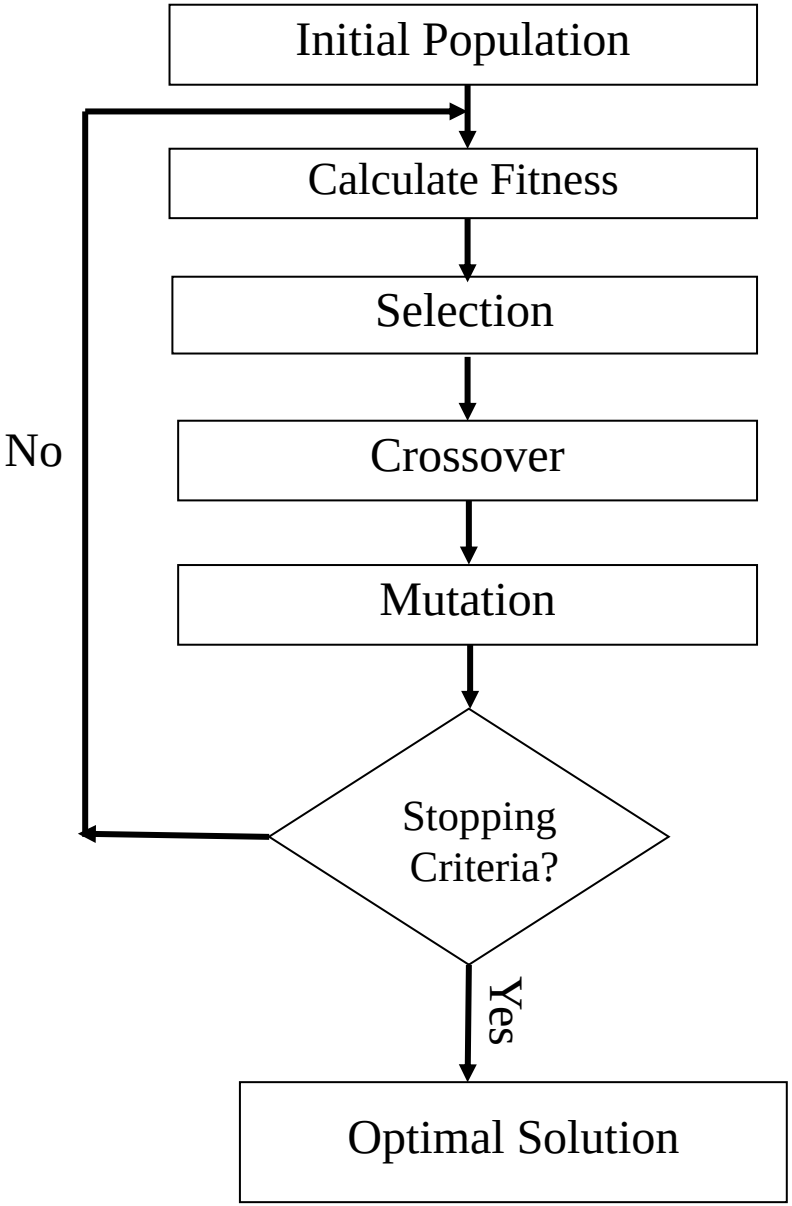
Keywords Related to GA (cont.)

- **Fitness Function**: A mathematical function used for calculating the fitness of a chromosome.
- **Crossover Probability**: A number between zero and one that indicates the probability of two chromosomes crossing over.
- **Mutation Probability**: A number between zero and one that indicates the probability of mutation occurring in a single gene.
- **Offspring**: An individual that was produced through reproduction. It also referred to as child.
- **Population**: A group of individuals that breed together.

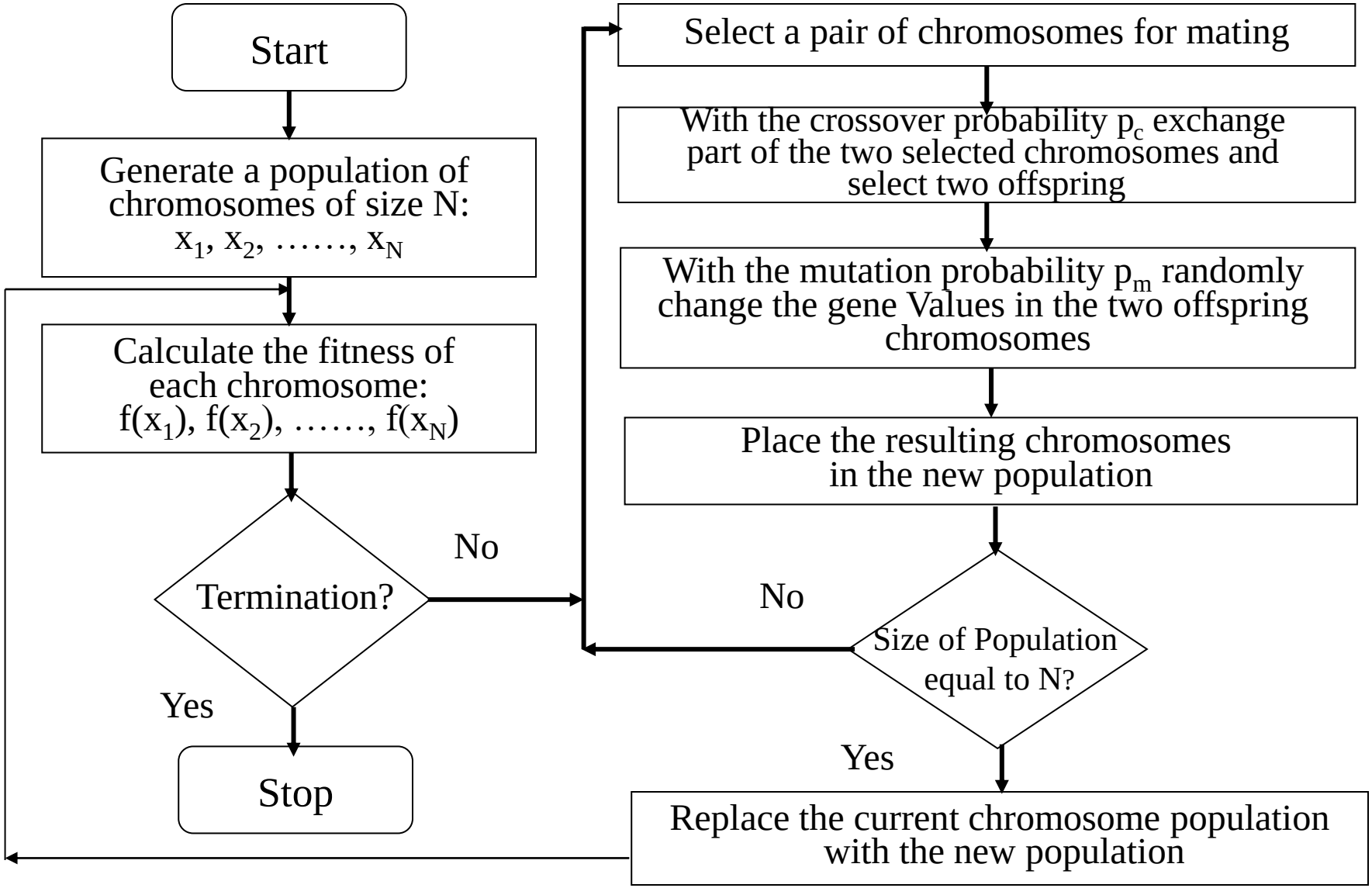
Genetic Algorithm(John Holland)

- ❖Abstraction of real biological evolution
- ❖Solves complex problems (like NP-hard problems)
- ❖Focus on optimization
- ❖Uses a population of possible/candidate solutions for a given problem
- ❖From a group of individuals, the best will survive (Survival of the Fittest)

Genetic Algorithm Flowchart



Genetic Algorithm Flowchart



Example Problem & Solution

- Find the maximum value of the function ($15x-x^2$), where parameter x varies between 0 and 15.
  - Chromosome can be built with only four gene (0-15)

Solution:

Suppose,

- The size of the chromosome population, N = 6
- The crossover probability, pc = 0.7
- The mutation probability, pm = 0.001
- The fitness function, $f(x) = 15x-x^2$

Example Problem & Solution

Step 1: GA creates an initial population of chromosomes by filling six 4-bit string with randomly generated ones and zeros

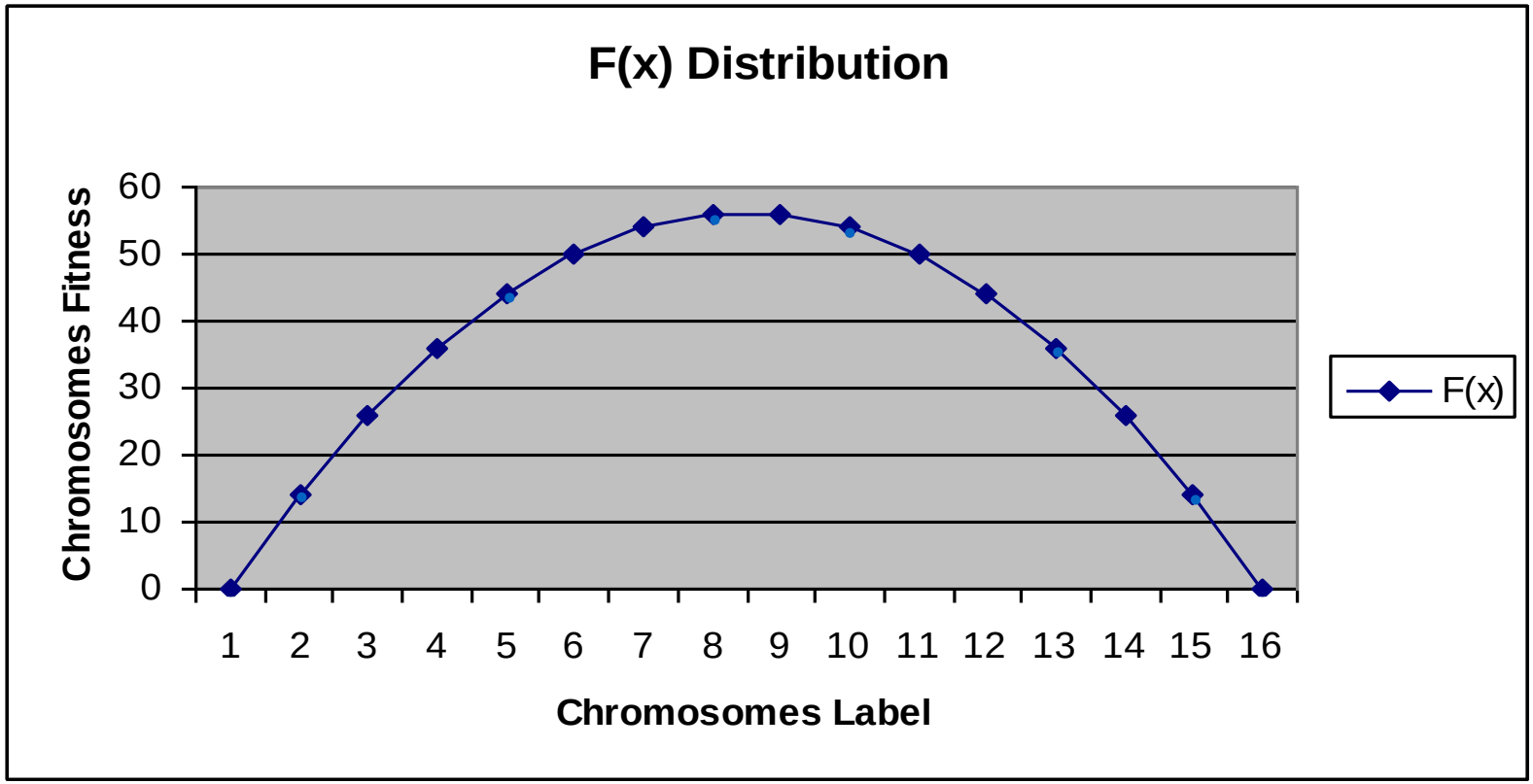
Chromosome Label	Chromosome String	Decoded Integer	Chromosome Fitness	Fitness Ratio
X1	1100	12		
X2	0100	4		
X3	0001	1		
X4	1110	14		
X5	0111	7		
X6	1001	9		

Example Problem & Solution

Step 2: Calculate the fitness of each individual Chromosome

Chromosome Label	Chromosome String	Decoded Integer	Chromosome Fitness	Fitness Ratio
X1	1100	12	36	
X2	0100	4	44	
X3	0001	1	14	
X4	1110	14	14	
X5	0111	7	56	
X6	1001	9	54	

Fitness Function and Chromosome Locations



Example Problem & Solution

Step 3: The fitness ratio determines the chromosome's chance of being selected for mating.

❖X5 & X6 have fair chance; X3 & X4 have less chance.

Chromosome Label	Chromosome String	Decoded Integer	Chromosome Fitness	Fitness Ratio
X1	1100	12	36	16.51
X2	0100	4	44	20.18
X3	0001	1	14	**6.42**
X4	1110	14	14	**6.42**
X5	0111	7	56	**25.68**
X6	1001	9	54	**24.77**

Fitness Function in Roulette Wheel Distribution

1100 (12)=36

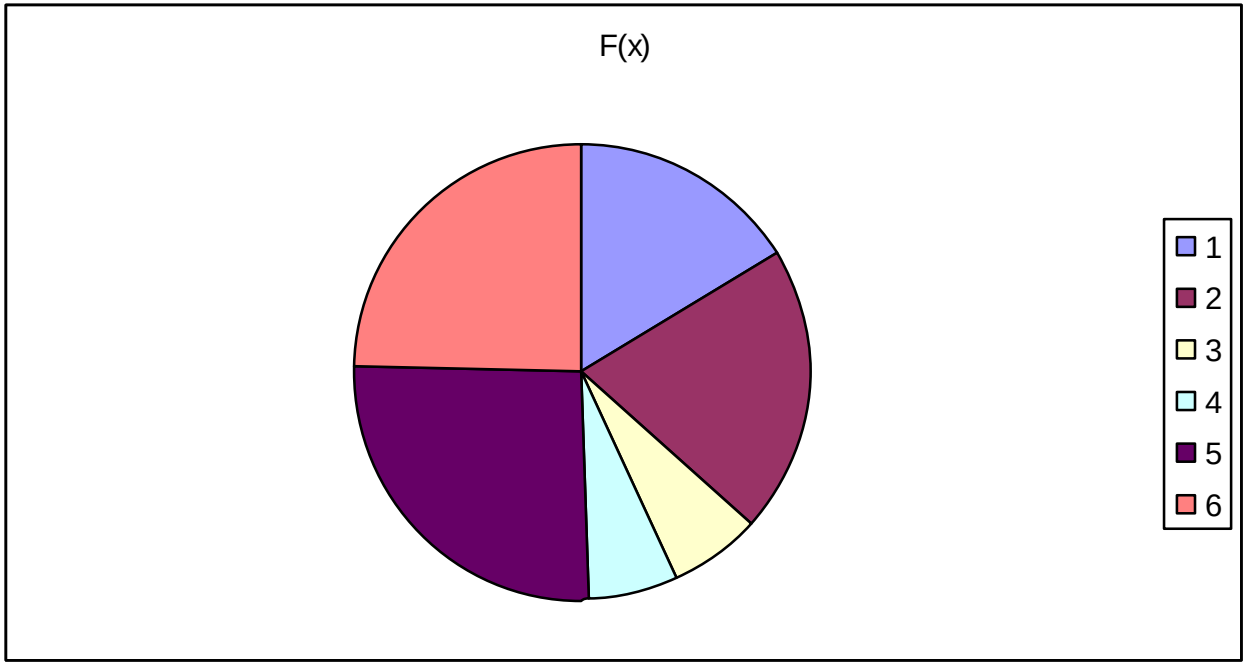
0100 (4)=44

0001 (1)=14

1110 (14)=14

0111 (7)=56

1001 (9)=54



Example Problem & Solution (Mating)

Step 4: Select chromosome for mating.

❖First X6 & X2 become parents, 2nd X1 & X5, then X2 & X5.

Generation i		Crossover
Chromosome Label	Chromosome String	
$X1_i$	1100	$X6_i$ 10-*01* $X2_i$ 01-*00*
$X2_i$	0100	
$X3_i$	0001	$X1_i$ 11-*00* $X5_i$ 01-*11*
$X4_i$	1110	
$X5_i$	0111	$X2_i$ 01-*00* $X5_i$ 01-*11*
$X6_i$	1001	

Example Problem & Solution (Crossover)

Step 5: Crossover Operator randomly chooses a crossover point where two parent chromosomes 'Break' and then exchange the chromosomes after that point.

Generation i		**Crossover**
Chromosome Label	Chromosome String	
$X1_i$	1100	$X6_i$ 10-*01* $X2_i$ 01-*00*
$X2_i$	0100	
$X3_i$	0001	$X1_i$ 11-*00* $X5_i$ 01-*11*
$X4_i$	1110	
$X5_i$	0111	$X2_i$ 01-*00* $X5_i$ 01-*11*
$X6_i$	1001	

Example Problem & Solution (Crossover)

Step 5: If a pair of chromosomes does not cross over then chromosome cloning take places and the offspring are exact copies of each parent.

Generation i		Crossover	
Chromosome Label	Chromosome String		
$X1_i$	1100	$X6_i$ 10-*01* $X2_i$ 01-*00*	$X6`_i$ 10*00*
$X2_i$	0100		$X2`_i$ 01**01**
$X3_i$	0001	$X1_i$ 11-*00* $X5_i$ 01-*11*	$X1`_i$ 1111
$X4_i$	1110		$X5`_i$ 01**00**
$X5_i$	0111	$X2_i$ 01-*00* $X5_i$ 01-*11*	$X5`_i$ 0100
$X6_i$	1001		$X2`_i$ 0111

Example Problem & Solution (Mutation)

Step 6: The mutation operator flips a randomly selected gene in a chromosome.

Generation i		**Crossover**	Mutation
Chromosome Label	Chromosome String		
$X1_i$	1100		$X6`_i$ 10*00*
$X2_i$	0100	$X6_i$ 10-*01* $X2_i$ 01-*00*	$X2`_i$ 01**01**
$X3_i$	0001		$X1`_i$ 1111, $X1``_i$ 1011
$X4_i$	1110	$X1_i$ 11-*00* $X5_i$ 01-*11*	$X5`_i$ 0100
$X5_i$	0111		$X2`_i$ 0100, $X2``_i$ 0110
$X6_i$	1001	$X2_i$ 01-*00* $X5_i$ 01-*11*	$X5`_i$ 0111

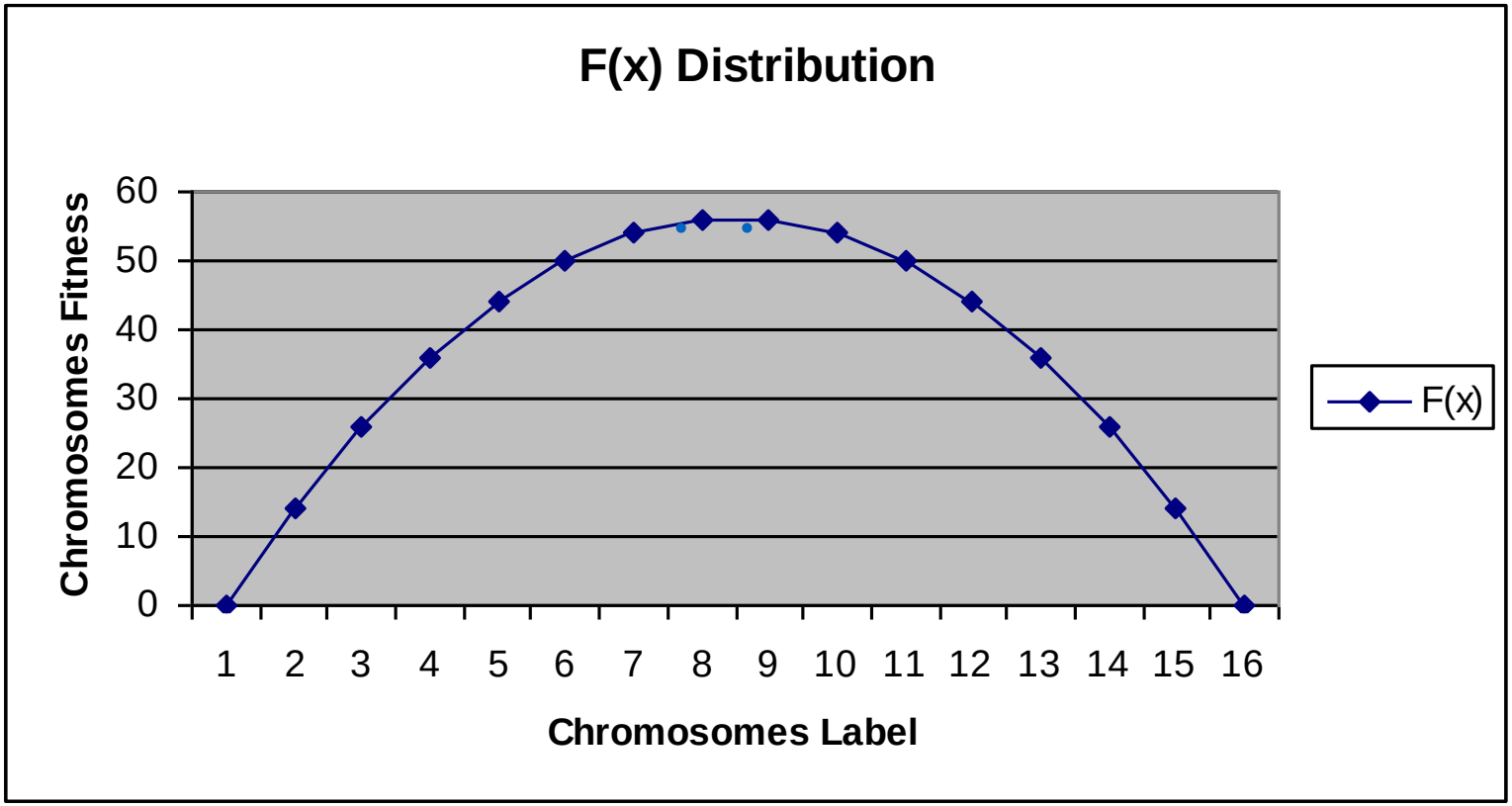
Example Problem & Solution

Step 7: The fitness ratio determines the chromosome's chance of being selected for mating.

❖X5 & X6 have fair chance; X3 & X4 have less chance.

Generation i		Generation i+1		
Chromosome Label	Chromosome String	Chromosome Label	Chromosome String	Fitness 15x-x2
$X1_i$	1100	$X1_{i+1}$	1000	**56**
$X2_i$	0100	$X2_{i+1}$	0101	50
$X3_i$	0001	$X3_{i+1}$	1011	44
$X4_i$	1110	$X4_{i+1}$	0100	44
$X5_i$	0111	$X5_{i+1}$	0110	54
$X6_i$	1001	$X6_{i+1}$	0111	**56**

Fitness Function and Chromosome Locations



Example Problem & Solution

Question

Generate 3 chromosomes, each containing **12 bits**. Apply **single-point** and **two-point crossover**. Apply **mutation** by generating a random number $r \in [0,1]$, where $r < p, p = 1/2 = 0.0833$

Solution

Initial Chromosomes:

C1 → 001001000111

C2 → 101101000011

C3 → 110110010001

Example Problem & Solution

Solution

Single-Point Crossover (at point 6)

C1 → 001001 | 000111

C2 → 101101 | 000011

Offspring

Ch1 → 001001000011

Ch2 → 101101000111

Example Problem & Solution

<u>Solution</u>

<u>Two-Point Crossover (2, 6)</u>

C1 → 00 | 1001 | 000111

C2 → 10 | 1101 | 000011

<u>Offspring</u>

Ch1 → 001101000111

Ch2 → 101001000011

Example Problem & Solution

Solution

Mutation on C3

- Mutation probability, $p = 1/2 = 0.0833$
- A bit is mutated when the generated random number satisfies: $r < p$

C3 with random numbers

Bit Position	1	2	3	4	5	6	7	8	9	10	11	12
C3 bit	1	1	0	1	1	0	0	1	0	0	0	1
Random no.	0.5	0.6	0.02	0.9	0.01	0.33	0.5	0.62	0.77	0.03	0.8	0.87

Example Problem & Solution

Solution

Mutation on C3

Mutated positions: Since these values are less than 0.0833, mutation occurs at:

Position 3 → 0.02

Position 5 → 0.01

Position 10 → 0.03

So the bits flip as:

Position 3: 0 → 1

Position 5: 1 → 0

Position 10: 0 → 1

Mutated Chromosome

C3' → 111100010101

Example Problem & Solution

Question

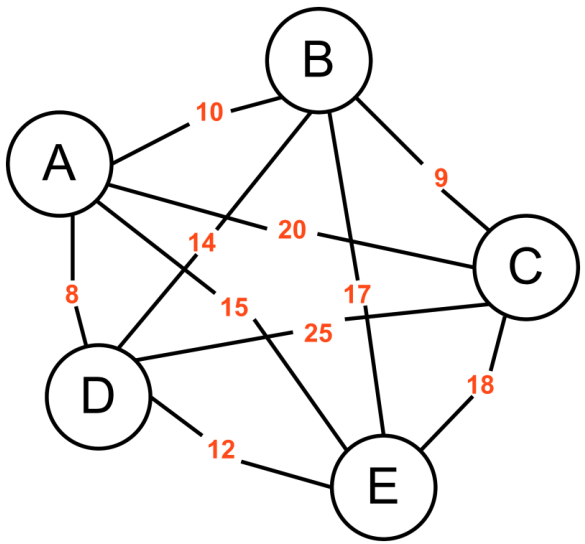
Travelling Salesman Problem by Genetic Algorithm

From Given Graph

(a) Choose **3 chromosomes** and find their **fitness values**.

(b) Apply **crossover** on chromosomes **ABCDE** and **ADBEC** at **single point = 2**.

(c) Apply **mutation** on chromosome: **ABDEC**



Example Problem & Solution

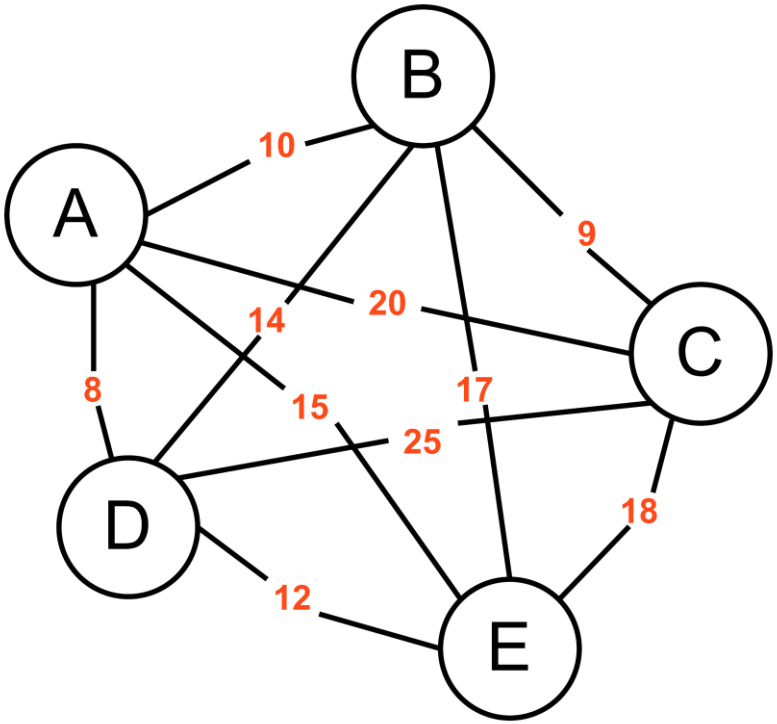
<u>Solution</u>

(a) Chosen Chromosomes

C1 → ABCDE

C2 → ABDEC

C3 → ADBEC



Fitness Function, $F = \frac{1}{D}$

- **For C1=ABCDE**

$$D1 = 10 + 9 + 25 + 12 = 56$$

$$F1 = \frac{1}{D1}$$

$$= \frac{1}{56}$$

Example Problem & Solution

<u>Solution</u>

- $\boldsymbol{For\ C2 = ABDEC}$

$$D2 = 10 + 14 + 12 + 18 = 54$$

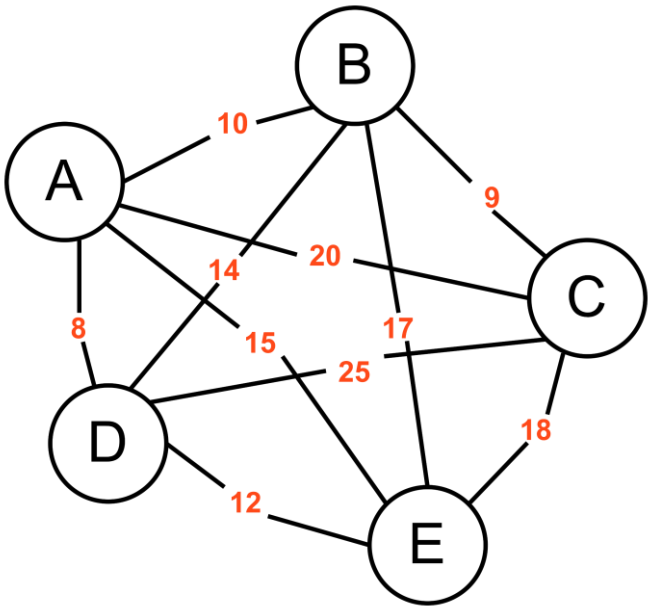
$$F2 = \frac{1}{D2} = \frac{1}{54}$$

- $\boldsymbol{For\ C3 = ADBEC}$

$$D3 = 8 + 14 + 17 + 18 = 57$$

$$F3 = \frac{1}{D3} = \frac{1}{57}$$

- **Final Fitness Values**
  - $\boldsymbol{C1 = ABCDE \rightarrow \frac{1}{56}}$
  - $\boldsymbol{C2 = ABDEC \rightarrow \frac{1}{54}}$
  - $\boldsymbol{C3 = ADBEC \rightarrow \frac{1}{57}}$



Example Problem & Solution

Solution

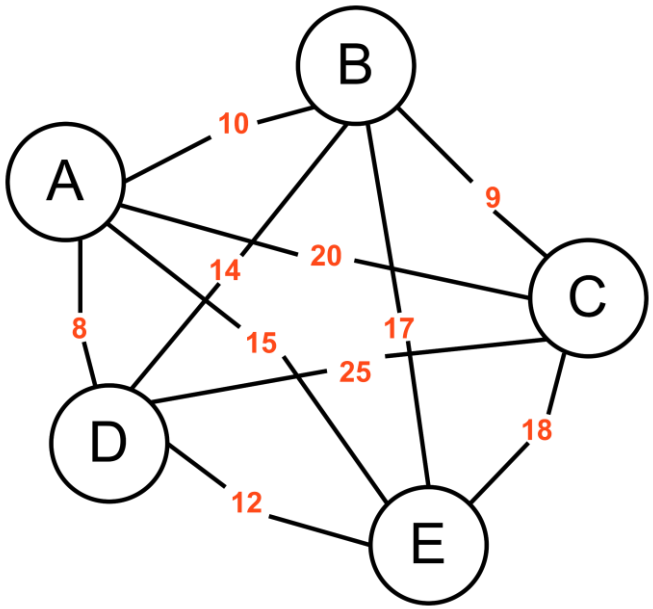
(b) Crossover

Parents

- **P1 → AB | CDE**
- **P2 → AD | BEC**

Children

- **Ch1 → ABDEC**
- **Ch2 → ADBCE**



(c) Mutation

Before Mutation chromosome shown is:

ABDEC

After Mutation chromosome shown is:

A E D B C

Example Problem & Solution

Question

Knapsack Problem by Genetic Algorithm

Given Data:

Knapsack Capacity

- Knapsack (K) = 12 kg

Chromosome

- A B C D

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Tasks

(i) Initialize **4 chromosomes** and show whether they are **accepted** or **rejected** by the knapsack. Calculate the **probability of selection** for **generation 2**.

(ii) Select chromosomes for **generation 2**, take **single-point crossover at point 3**, and generate **generation 3**. Show whether the chromosomes are **accepted** or **rejected** by the knapsack and **comment on the result**.

Example Problem & Solution

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Solution

Example Chromosome

- Chromosome: 1010

This means **A** and **C** are selected.

- Weight = 5+7 = 12 kg
- Since 12≤12 , it is **Accepted**

(i) Initial 4 Chromosomes (Generation-1)

- $\boldsymbol{C1} \rightarrow 0101$
- $\boldsymbol{C2} \rightarrow 0110$
- $\boldsymbol{C3} \rightarrow 1101$
- $\boldsymbol{C4} \rightarrow 1111$

C1 = 0101

Selected items: **B, D**

- $Weight = 3 + 2 = 5\ kg < 12\ kg$
- $Value = 5 + 7 = 12$

So, **C1 is Accepted**

Fitness (C1)=12

Example Problem & Solution

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Solution

C2 = 0110

Selected items: **B, C**

- $Weight = 3 + 7 = 10\ kg < 12\ kg$
- $Value = 5 + 10 = 15$

So, **C2 is Accepted**

F(C2)=15

C3 = 1101

Selected items: **A, B, D**

- $Weight = 5 + 3 + 2 = 10\ kg < 12\ kg$
- $Value = 12 + 5 + 7 = 24$

So, **C3 is Accepted**

F(C3)=24

Example Problem & Solution

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Solution

C4 = 1111

Selected items: **A, B, C, D**

- Weight =5+3+7+2=17 kg>12 kg

So, **C4 is Rejected**

F(C4)=0

Selection Probability for Generation 2

Total fitness:

$$12 + 15 + 24 + 0 = 51$$

So the probabilities are:

- $P(C1) = \frac{12}{51} = 0.2353$
- $P(C2) = \frac{15}{51} = 0.2941$
- $P(C3) = \frac{24}{51} = 0.4706$
- $P(C4) = \frac{0}{51} = 0$

Example Problem & Solution

Solution

Fitness Table

Chromosome	Fitness Value (F)	Probability for Next generation (Pi)
C1	12	0.2353
C2	15	0.2941
C3	24	0.4706
C4	0	0
	ΣF=51 (Gen -1)	

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Example Problem & Solution

Solution

(ii) Roulette Wheel Selection

Generation 2

Selected chromosomes:

- C3 → 1101
- C2 → 0110
- C3 → 1101
- C1 → 0101

Single-Point Crossover at Point 3

Parents:

- **P3** → 110|1
- **P2** → 011|0

Offspring:

- **OS-1** → 1100
- **OS-2** → 0111

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Example Problem & Solution

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Solution

Generation 3

Chromosomes in generation 3:

- OS-1 → 1100
- OS-2 → 0111
- C3 → 1101
- C1 → 0101

Checking Acceptance / Rejection

OS-1 = 1100

Selected items: A, B

- $Weight = 5 + 3 = 8\ kg < 12\ kg$
- $Value = 12 + 5 = 17$

Accepted

Example Problem & Solution

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Solution

OS-2 = 0111

Selected items: **B, C, D**

- $Weight = 3 + 7 + 2 = 12\ kg \leq 12\ kg$
- $Value = 5 + 10 + 7 = 22$

Accepted

C3 = 1101

Selected items: **A, B, D**

- $Weight = 5 + 3 + 2 = 10\ kg < 12\ kg$
- $Value = 12 + 5 + 7 = 24$

Accepted

C1 = 0101

Selected items: **B, D**

- $Weight = 3 + 2 = 5\ kg < 12\ kg$
- $Value = 5 + 7 = 12$

Accepted

Example Problem & Solution

Solution

Fitness Table for Generation 3

Chromosome	Fitness
OS-1	17
OS-2	22
C3	24
C1	12

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Comment on the Result

- All chromosomes in **generation 3** are **accepted** by the knapsack. The **best chromosome** is still:
  - **C3 = 1101**
  - **Fitness = 24**
- So, crossover produced valid offspring, but it did **not outperform C3**.

Example Problem & Solution

<u>Solution</u>

Generation 3 Fitness Summary

Chromosome	Fitness
OS1	$17
OS2	$22
C3	$24
C1	$12

ΣF3=75

- $\Sigma F1 = 51$
- $\Sigma F3 = 75$

Item	Weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Example Problem

Question

Warehouse Problem by Genetic Algorithm

- Consider a big businessman has **4 warehouses** and he supplies his **goods** to **4 markets**. The distance in **KM** of each market from each warehouse is given in the table below. Each warehouse supplies goods to **one market**, i.e., the demand of each market is satisfied by **one warehouse**.

Distance Table

Warehouse	Market-1	Market-2	Market-3	Market-4
Wh-1	23	27	17	23
Wh-2	15	10	14	8
Wh-3	11	14	25	32
Wh-4	9	22	13	8

Example Problem

Question

Tasks

(i) Generate **4 initial population**.

(ii) Apply **crossover** between two chromosomes using **random number thresholding**.

(iii) Apply **crossover again** then find the result of **iteration-3**.

(iv) Comment on the result of **iteration-2 and 3**.

Applications of Genetic Algorithm

- Links:
- https://www.quora.com/What-are-some-real-world-applications-of-genetic-algorithms
- http://brainz.org/15-real-world-applications-genetic-algorithms/
- https://en.wikipedia.org/wiki/List_of_genetic_algorithm_applications

Recommended Textbooks

- [Negnevitsky, 2001] M. Negnevitsky " Artificial Intelligence: A guide to Intelligent Systems", Pearson Education Limited, England, 2002.
- [Russell, 2003] S. Russell and P. Norvig Artificial Intelligence: A Modern Approach Prentice Hall, 2003, Second Edition
- [Patterson, 1990] D. W. Patterson, "Introduction to Artificial Intelligence and Expert Systems", Prentice-Hall Inc., Englewood Cliffs, N.J, USA, 1990.
- [Lindsay, 1997] P. H. Lindsay and D. A. Norman, Human Information Processing: An Introduction to Psychology, Academic Press, 1977.